

Tutorial 2

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Date: 12th August 2026

1 Regular Expressions

Q1. Write regular expressions for the following

1. The set of all strings over $\{0, 1\}$ such that every pair of adjacent 0s appears before any pair of adjacent 1s. That is, no substring 11 occurs before a substring 00.
2. The set of all strings over $\{a, b\}$ that contain exactly two occurrences of the substring ab .
3. The set of all strings over $\{0, 1\}$ that contain an even number of 0s and an odd number of 1s.
4. The set of all strings over $\{a, b\}$ that contain aba as a substring but do not contain bab as a substring.

Q2. Describe the languages represented by the following regular expressions. Then, determine whether the two regular expressions define the same language. Justify your answer.

1. $r_1 = (01)^* + (10)^* + 0(10)^* + 1(01)^*$
2. $r_2 = (\varepsilon + 1)(01)^*(\varepsilon + 0)$

2 Pumping Lemma

Q1. State whether the following statements are true or false and justify your answer.

1. If L is non-regular, then L does not satisfy the Pumping Lemma.
2. If L is non-regular, then every pumping length fails.
3. If L is regular, then every sufficiently long string in L can be pumped.

Q2. Are the following languages regular? If not disprove using pumping lemma, otherwise, give a finite automaton $A : \mathcal{L}(A) = \mathcal{L}$ and a regular expression

1. $\mathcal{L} = \{w \in \{(,)\}^* \mid w \text{ is a balanced string of parentheses}\}$.
2. $\mathcal{L} = \{a^i b^j \mid \gcd(i, j) = 1\}$:
3. $\mathcal{L} = \{w \in \{a, b, c\}^* \mid |w| = n^2 \text{ for some } n \in \mathbb{N}\}$.
4. $\mathcal{L} = \{w \in \{0, 1\}^* \mid w \text{ is a palindrome}\}$
5. $\mathcal{L} = \{0^n : n \text{ is a Fibonacci/Pingala Number}\}$

Q3. Give an example of a regular language and a non-regular language that satisfies the pumping lemma.

Q4. Consider language $\mathcal{L}_1 \subseteq \{0\}^*$ that is infinite and regular. Prove that $\exists c > 0$ such that there are infinitely many pairs of strings in $\mathcal{L}_1$ whose lengths differ by *at most* c .Use this to prove that $\mathcal{L}_2 = \{0^{n!} \mid n \geq 0\}$ is not regular.

Q5. Consider $\mathcal{L} = \{0^i 1^j \mid i < j\}$. The pumping lemma says that there exists a pumping length p . You choose a string $w \in \mathcal{L}$ with $|w| \geq p$. Your opponent chooses a valid decomposition $w = xyz$. You win if you can choose $k \geq 0$ such that $xy^k z \notin \mathcal{L}$, regardless of the decomposition your opponent chooses. Find a winning strategy and justify why it works. Also what can you conclude about the language?

Q6. Consider the language

$$F = \{a^i b^j c^k \mid i = 1 \Rightarrow j = k\}.$$

1. Show that the ordinary Pumping Lemma is insufficient to prove that F is non-regular. Explain why, for every sufficiently long $w \in F$, there may exist a valid decomposition whose pumping preserves membership.
2. Use the Subword Pumping Lemma to prove that F is not regular.
3. Explain precisely what restriction in the ordinary Pumping Lemma prevents it from detecting the non-regularity of F , and how the Subword Pumping Lemma removes this restriction.